

Weight Formula Triangle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The triangle links weight (W), mass (m), and gravity (g). Cover the unknown to read the formula: $W = m \times g$, $m = W \div g$, and $g = W \div m$. Use $g = 10 \text{ N/kg}$.

Round 1 — Retrieve It

1. What three quantities does the weight triangle connect?

2. Write the formula for weight.

3. Write the formula for mass.

4. Which variable do you cover to find gravitational field strength (g), and what formula results?

Round 2 — Sketch It

Draw the weight triangle. Put weight on top; put mass and gravity on the bottom. Circle the two quantities you would multiply to find weight.

Weight — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed. Use $g = 10 \text{ N/kg}$.

Round 3 — Apply the Relationship

1. A backpack has a mass of 5 kg. What is its weight? ($g = 10 \text{ N/kg}$)

2. A crate has a weight of 90 N. What is its mass? ($g = 10 \text{ N/kg}$)

3. An object with a mass of 4 kg has a weight of 60 N. What is the gravitational field strength (g)?

Round 4 — Reason from Evidence

A rock has a mass of 5 kg. On Earth $g = 10 \text{ N/kg}$; on the Moon $g \approx 1.6 \text{ N/kg}$. Use the triangle to find the rock's weight in each place, then explain why its weight changes but its mass does not.

Weight — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain how the weight triangle helps you find a missing quantity. Use one example that is not printed on the cue card.

C

Connect

Connect weight to mass. Mass is the amount of matter in an object and stays the same everywhere; weight is the pull of gravity and depends on g . Explain why an astronaut's mass is unchanged on the Moon but their weight is smaller.

E

Extend

Create your own problem that asks for mass. Give the numbers and show the work using $m = W \div g$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Weight

Remove this page before student copying. Answers below give expected ideas, not required wording. Use $g = 10 \text{ N/kg}$ throughout.

Retrieve It

1. Weight, mass, and gravitational field strength (gravity).
2. $W = m \times g$.
3. $m = W \div g$.
4. Cover g ; $g = W \div m$.

Sketch It — look for

Weight on top, with mass and gravity on the bottom. Weight is found by multiplying mass $\times$ gravity (the two bottom quantities).

Apply It

1. $W = m \times g = 5 \times 10 = \mathbf{50 \text{ N}}$.
2. $m = W \div g = 90 \div 10 = \mathbf{9 \text{ kg}}$.
3. $g = W \div m = 60 \div 4 = \mathbf{15 \text{ N/kg}}$.

Reason from Evidence — look for

Earth: $W = m \times g = 5 \times 10 = 50 \text{ N}$. Moon: $W = m \times g = 5 \times 1.6 = 8 \text{ N}$. The weight is smaller on the Moon because g is smaller there, but the mass stays 5 kg because mass is the amount of matter and does not depend on gravity.

ACE — Extend look-for

Many answers possible. Check that the problem asks for mass and the work correctly uses $m = W \div g$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.